

Logistics & Transportation

Fleet Performance & Delivery Efficiency

Data Cleaning & Modeling

Remove NULL Rows & Columns

Queries [2] × ✓ f_x Table.RemoveColumns(#"Changed Type",{"Column10", "Column11", "Column12", "Column13", "Column14", "Column15",

	Destination	Distance_km	Fuel_Consumed_L	Delivery_Status	Delivery_Date
1	Pune	1173	108.42	On-Time	27-01-2023 00:00:00
2	Bangalore	1727	161.33	On-Time	21-02-2023 00:00:00
3	Pune	1459	154.7	On-Time	17-02-2023 00:00:00
4	Pune	382	26.6	On-Time	18-02-2023 00:00:00
5	Mumbai	398	33.2	On-Time	15-02-2023 00:00:00
6	Mumbai	1275	85.04	Late	25-02-2023 00:00:00
7	Kolkata	752	58.08	On-Time	19-01-2023 00:00:00
8	Pune	74	5.24	On-Time	01-01-2023 00:00:00
9	Hyderabad	186	16.22	On-Time	23-02-2023 00:00:00
10	Hyderabad	1375	105.21	Late	02-02-2023 00:00:00
11	Hyderabad	419	31.17	On-Time	21-01-2023 00:00:00
12	Delhi	751	51.77	On-Time	15-02-2023 00:00:00
13	Chennai	1571	188.52	Late	02-02-2023 00:00:00
14	Bangalore	1524	104.51	On-Time	16-02-2023 00:00:00

Query Settings ×

PROPERTIES

Name
Trip_Data

All Properties

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type
- Removed Columns**
- Removed Blank Rows
- Renamed Columns

Queries [2] × ✓ f_x Table.SelectRows(#"Removed Columns", each not List.IsEmpty(List.RemoveMatchingItems(Record.FieldValues(_), {"",

	Destination	Distance_km	Fuel_Consumed_L	Delivery_Status	Delivery_Date
1	Pune	1173	108.42	On-Time	27-01-2023 00:00:00
2	Bangalore	1727	161.33	On-Time	21-02-2023 00:00:00
3	Pune	1459	154.7	On-Time	17-02-2023 00:00:00
4	Pune	382	26.6	On-Time	18-02-2023 00:00:00
5	Mumbai	398	33.2	On-Time	15-02-2023 00:00:00
6	Mumbai	1275	85.04	Late	25-02-2023 00:00:00
7	Kolkata	752	58.08	On-Time	19-01-2023 00:00:00
8	Pune	74	5.24	On-Time	01-01-2023 00:00:00
9	Hyderabad	186	16.22	On-Time	23-02-2023 00:00:00
10	Hyderabad	1375	105.21	Late	02-02-2023 00:00:00
11	Hyderabad	419	31.17	On-Time	21-01-2023 00:00:00
12	Delhi	751	51.77	On-Time	15-02-2023 00:00:00
13	Chennai	1571	188.52	Late	02-02-2023 00:00:00
14	Bangalore	1524	104.51	On-Time	16-02-2023 00:00:00

Query Settings ×

PROPERTIES

Name
Trip_Data

All Properties

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type
- Removed Columns
- Removed Blank Rows**
- Renamed Columns

Queries [2] × ✓ f_x Table.SelectRows(#"Changed Type", each not List.IsEmpty(List.RemoveMatchingItems(Record.FieldValues(_), {"",

	Vehicle_ID	Vehicle_Type	Capacity_kg	Maintenance_Cost
1	V01	Van	8424	5633
2	V02	Truck	1970	6776
3	V03	Truck	1207	18031
4	V04	Mini-Truck	8803	9033
5	V05	Truck	9941	17751
6	V06	Van	6053	5914
7	V07	Van	2598	12837

Query Settings ×

PROPERTIES

Name
Vehicle_Master

All Properties

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type
- Removed Blank Rows**

Split Column

Group By Use First Row as Headers Count Rows Reverse Rows Detect Data Type Fill Move Split Format ABC Extract Statistics Standard Scientific Rounding Date Time Duration Run R script Run Python script

Queries [2] × ✓ f_x Table

	Destination
1	Pune
2	Bangalore
3	Pune
4	Pune
5	Mumbai
6	Mumbai
7	Kolkata
8	Pune
9	Hyderabad
10	Hyderabad
11	Hyderabad
12	Delhi
13	Chennai
14	Bangalore
15	Mumbai
16	Mumbai
17	Kolkata
18	Delhi

Split Column by Delimiter

Specify the delimiter used to split the text column.

Select or enter delimiter
Space

Split at
☐ Left-most delimiter
☐ Right-most delimiter
☒ Each occurrence of the delimiter

Advanced options
Quote Character
"
☐ Split using special characters
Insert special character

OK Cancel

Query Settings ×

PROPERTIES

Name
Trip_Data

All Properties

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type
- Removed Columns
- Removed Blank Rows
- Renamed Columns
- Replaced Value
- Replaced Value1
- Replaced Value2
- Replaced Value3
- Replaced Value4
- Replaced Value5
- Replaced Value6

Queries [2] × ✓ f_x Table.TransformColumnTypes(#"Split Column by Delimiter",{{"Delivery_Date.1", type date}, {"Delivery_Date.2", type date}} Query Settings ×

	Distance_km	Fuel_Consumed_Liters	Delivery_Status	Delivery_Date.1	Delivery_Date.2
1	1173	108.42	On-Time	27-01-2023	00:00:00
2	1727	161.33	On-Time	21-02-2023	00:00:00
3	1459	154.7	On-Time	17-02-2023	00:00:00
4	382	26.6	On-Time	18-02-2023	00:00:00
5	398	33.2	On-Time	15-02-2023	00:00:00
6	1275	85.04	Late	25-02-2023	00:00:00
7	752	58.08	On-Time	19-01-2023	00:00:00
8	74	5.24	On-Time	01-01-2023	00:00:00
9	186	16.22	On-Time	23-02-2023	00:00:00
10	1375	105.21	Late	02-02-2023	00:00:00
11	419	31.17	On-Time	21-01-2023	00:00:00
12	751	51.77	On-Time	15-02-2023	00:00:00
13	1571	188.52	Late	02-02-2023	00:00:00
14	1524	104.51	On-Time	16-02-2023	00:00:00

PROPERTIES
Name: Trip_Data

APPLIED STEPS
Navigation
Promoted Headers
Changed Type
Removed Columns
Removed Blank Rows
Renamed Columns
Replaced Value
Replaced Value1
Replaced Value2
Replaced Value3

Remove Column

Queries [2] × ✓ f_x Table.RemoveColumns(#"Changed Type1",{"Delivery_Date.2"}) Query Settings ×

	Destination	Distance_km	Fuel_Consumed_Liters	Delivery_Status	Delivery_Date.1
1	Pune	1173	108.42	On-Time	27-01-2023
2	Bangalore	1727	161.33	On-Time	21-02-2023
3	Pune	1459	154.7	On-Time	17-02-2023
4	Pune	382	26.6	On-Time	18-02-2023
5	Mumbai	398	33.2	On-Time	15-02-2023
6	Mumbai	1275	85.04	Late	25-02-2023
7	Kolkata	752	58.08	On-Time	19-01-2023
8	Pune	74	5.24	On-Time	01-01-2023
9	Hyderabad	186	16.22	On-Time	23-02-2023
10	Hyderabad	1375	105.21	Late	02-02-2023
11	Hyderabad	419	31.17	On-Time	21-01-2023
12	Delhi	751	51.77	On-Time	15-02-2023
13	Chennai	1571	188.52	Late	02-02-2023
14	Bangalore	1524	104.51	On-Time	16-02-2023

PROPERTIES
Name: Trip_Data

APPLIED STEPS
Promoted Headers
Changed Type
Removed Columns
Removed Blank Rows
Renamed Columns
Replaced Value
Replaced Value1
Replaced Value2
Replaced Value3
Replaced Value4

Replace Values

Queries [2] × ✓ f_x Table.ReplaceValue(#"Replaced Value4", "Built a dashboard to analyze on-time deliveries, fuel efficiency, and", ...) Query Settings ×

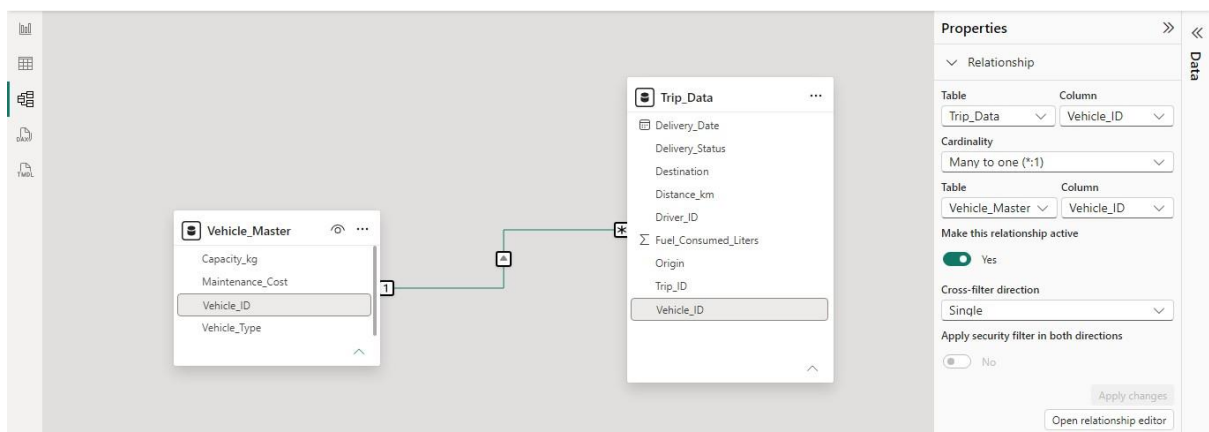
	Trip_ID	Vehicle_ID	Driver_ID	Origin	Destination	Distance_km
1	Unknown					
2	Unknown					
3	Unknown					
4	Unknown					
5	Unknown					
6	Unknown					
7	Unknown					
8	Tools Used: Power BI, D					
9	Unknown					
10	Bike Buyers Data Analy					
11	Cleaned and transforme					
12	Tools Used: Excel, Pivot					
13	T013					
14	T014					

Replace Values
Replace one value with another in the selected columns.
Value To Find: Tools Used: Power BI, DAX, Excel
Replace With: Unknown

PROPERTIES
Name: Trip_Data

APPLIED STEPS
Source
Navigation
Promoted Headers
Changed Type
Removed Columns
Removed Blank Rows
Renamed Columns
Replaced Value
Replaced Value1
Replaced Value2

Relate Trips Table with Vehicle Master Table



DAX Measures

Total Trips

1 Total Trips = COUNTROWS(Trip_Data)								
Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Fuel_Consumed_Liters	Delivery_Status	Delivery_Date	Route
V04	D01	Delhi	Pune	1173	108.42	On-Time	27 January 2023	Delhi-->Pune
V06	D08	Mumbai	Bangalore	1727	161.33	On-Time	21 February 2023	Mumbai-->Bangalore
V06	D08	Mumbai	Pune	1459	154.70	On-Time	17 February 2023	Mumbai-->Pune
V04	D09	Hyderabad	Pune	382	26.60	On-Time	18 February 2023	Hyderabad-->Pune
V06	D08	Pune	Mumbai	398	33.20	On-Time	15 February 2023	Pune-->Mumbai
V06	D07	Chennai	Mumbai	1275	85.04	Late	25 February 2023	Chennai-->Mumbai
V07	D03	Chennai	Kolkata	752	58.08	On-Time	19 January 2023	Chennai-->Kolkata
V02	D10	Delhi	Pune	74	5.24	On-Time	01 January 2023	Delhi-->Pune
V02	D07	Delhi	Hyderabad	186	16.22	On-Time	23 February 2023	Delhi-->Hyderabad
V02	D02	Bangalore	Hyderabad	1375	105.21	Late	02 February 2023	Bangalore-->Hyderabad
V06	D03	Kolkata	Hyderabad	419	31.17	On-Time	21 January 2023	Kolkata-->Hyderabad
V06	D01	Kolkata	Delhi	751	51.77	On-Time	15 February 2023	Kolkata-->Delhi
V05	D04	Kolkata	Chennai	1571	188.52	Late	02 February 2023	Kolkata-->Chennai
V05	D05	Hyderabad	Bangalore	1524	104.51	On-Time	16 February 2023	Hyderabad-->Bangalore
V05	D06	Kolkata	Mumbai	1956	179.88	On-Time	21 January 2023	Kolkata-->Mumbai

On-Time Delivery %

1 On-Time Delivery % = divide(calculate([Total Trips],Trip_Data[Delivery_Status]="on-time"),[Total Trips],0)								
Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Fuel_Consumed_Liters	Delivery_Status	Delivery_Date	Route
V04	D01	Delhi	Pune	1173	108.42	On-Time	27 January 2023	Delhi-->Pune
V06	D08	Mumbai	Bangalore	1727	161.33	On-Time	21 February 2023	Mumbai-->Bangalore
V06	D08	Mumbai	Pune	1459	154.70	On-Time	17 February 2023	Mumbai-->Pune
V04	D09	Hyderabad	Pune	382	26.60	On-Time	18 February 2023	Hyderabad-->Pune
V06	D08	Pune	Mumbai	398	33.20	On-Time	15 February 2023	Pune-->Mumbai
V06	D07	Chennai	Mumbai	1275	85.04	Late	25 February 2023	Chennai-->Mumbai
V07	D03	Chennai	Kolkata	752	58.08	On-Time	19 January 2023	Chennai-->Kolkata
V02	D10	Delhi	Pune	74	5.24	On-Time	01 January 2023	Delhi-->Pune
V02	D07	Delhi	Hyderabad	186	16.22	On-Time	23 February 2023	Delhi-->Hyderabad
V02	D02	Bangalore	Hyderabad	1375	105.21	Late	02 February 2023	Bangalore-->Hyderabad
V06	D03	Kolkata	Hyderabad	419	31.17	On-Time	21 January 2023	Kolkata-->Hyderabad
V06	D01	Kolkata	Delhi	751	51.77	On-Time	15 February 2023	Kolkata-->Delhi
V05	D04	Kolkata	Chennai	1571	188.52	Late	02 February 2023	Kolkata-->Chennai
V05	D05	Hyderabad	Bangalore	1524	104.51	On-Time	16 February 2023	Hyderabad-->Bangalore
V05	D06	Kolkata	Mumbai	1956	179.88	On-Time	21 January 2023	Kolkata-->Mumbai

Maintenance Cost Per KM

1 Maintenance Cost Per KM = divide(sum(Vehicle_Master[Maintenance_Cost]),sum(Trip_Data[Distance_km]))								
Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Fuel_Consumed_Liters	Delivery_Status	Delivery_Date	Route
V04	D01	Delhi	Pune	1173	108.42	On-Time	27 January 2023	Delhi-->Pune
V06	D08	Mumbai	Bangalore	1727	161.33	On-Time	21 February 2023	Mumbai-->Bangalore
V06	D08	Mumbai	Pune	1459	154.70	On-Time	17 February 2023	Mumbai-->Pune
V04	D09	Hyderabad	Pune	382	26.60	On-Time	18 February 2023	Hyderabad-->Pune
V06	D08	Pune	Mumbai	398	33.20	On-Time	15 February 2023	Pune-->Mumbai
V06	D07	Chennai	Mumbai	1275	85.04	Late	25 February 2023	Chennai-->Mumbai
V07	D03	Chennai	Kolkata	752	58.08	On-Time	19 January 2023	Chennai-->Kolkata
V02	D10	Delhi	Pune	74	5.24	On-Time	01 January 2023	Delhi-->Pune
V02	D07	Delhi	Hyderabad	186	16.22	On-Time	23 February 2023	Delhi-->Hyderabad
V02	D02	Bangalore	Hyderabad	1375	105.21	Late	02 February 2023	Bangalore-->Hyderabad
V06	D03	Kolkata	Hyderabad	419	31.17	On-Time	21 January 2023	Kolkata-->Hyderabad
V06	D01	Kolkata	Delhi	751	51.77	On-Time	15 February 2023	Kolkata-->Delhi
V05	D04	Kolkata	Chennai	1571	188.52	Late	02 February 2023	Kolkata-->Chennai
V05	D05	Hyderabad	Bangalore	1524	104.51	On-Time	16 February 2023	Hyderabad-->Bangalore
V05	D06	Kolkata	Mumbai	1956	179.88	On-Time	21 January 2023	Kolkata-->Mumbai

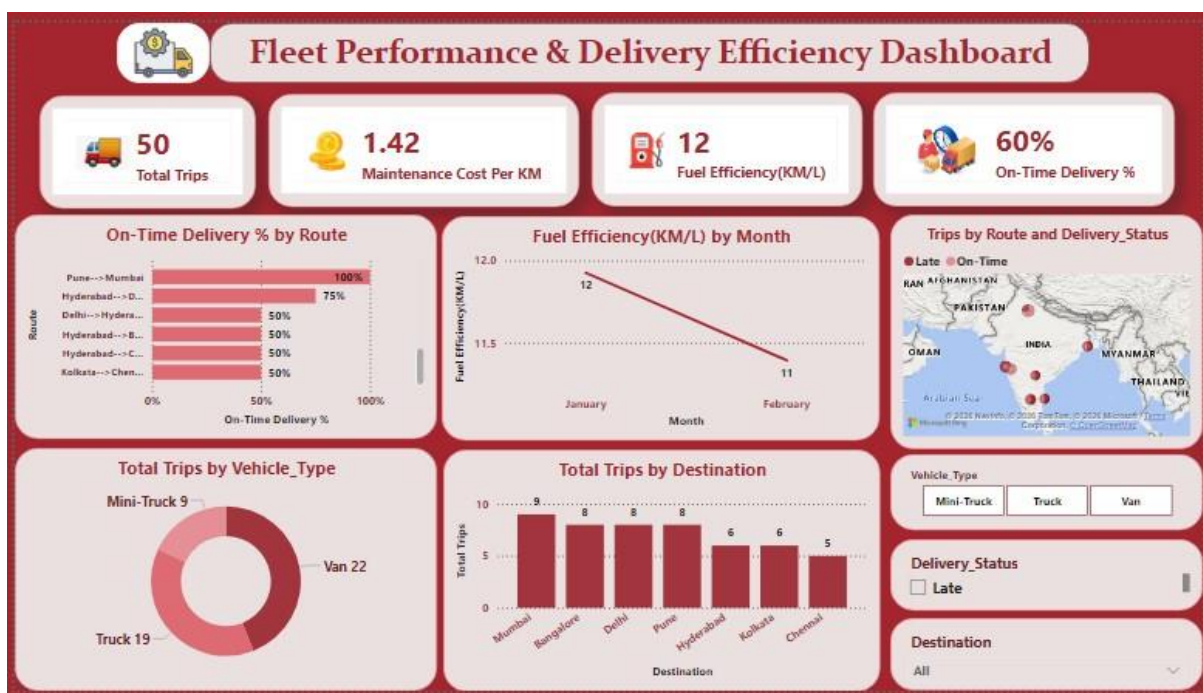
Fuel Efficiency(KM/L)

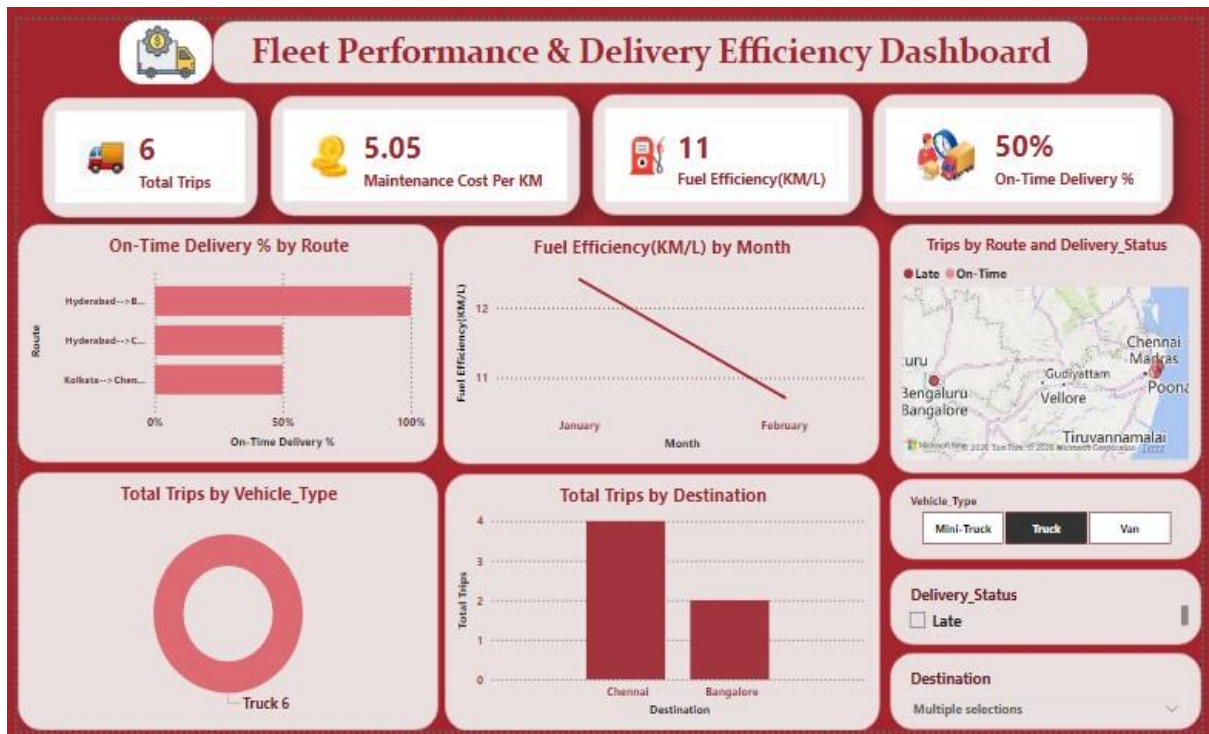
1 Fuel Efficiency(KM/L) = divide(sum(Trip_Data[Distance_km]),sum(Trip_Data[Fuel_Consumed_Liters]))								
Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Fuel_Consumed_Liters	Delivery_Status	Delivery_Date	Route
V04	D01	Delhi	Pune	1173	108.42	On-Time	27 January 2023	Delhi-->Pune
V06	D08	Mumbai	Bangalore	1727	161.33	On-Time	21 February 2023	Mumbai-->Bangalore
V06	D08	Mumbai	Pune	1459	154.70	On-Time	17 February 2023	Mumbai-->Pune
V04	D09	Hyderabad	Pune	382	26.60	On-Time	18 February 2023	Hyderabad-->Pune
V06	D08	Pune	Mumbai	398	33.20	On-Time	15 February 2023	Pune-->Mumbai
V06	D07	Chennai	Mumbai	1275	85.04	Late	25 February 2023	Chennai-->Mumbai
V07	D03	Chennai	Kolkata	752	58.08	On-Time	19 January 2023	Chennai-->Kolkata
V02	D10	Delhi	Pune	74	5.24	On-Time	01 January 2023	Delhi-->Pune
V02	D07	Delhi	Hyderabad	186	16.22	On-Time	23 February 2023	Delhi-->Hyderabad
V02	D02	Bangalore	Hyderabad	1375	105.21	Late	02 February 2023	Bangalore-->Hyderabad
V06	D03	Kolkata	Hyderabad	419	31.17	On-Time	21 January 2023	Kolkata-->Hyderabad
V06	D01	Kolkata	Delhi	751	51.77	On-Time	15 February 2023	Kolkata-->Delhi
V05	D04	Kolkata	Chennai	1571	188.52	Late	02 February 2023	Kolkata-->Chennai
V05	D05	Hyderabad	Bangalore	1524	104.51	On-Time	16 February 2023	Hyderabad-->Bangalore
V05	D06	Kolkata	Mumbai	1956	179.88	On-Time	21 January 2023	Kolkata-->Mumbai
V05	D06	Bangalore	Mumbai	858	92.70	Late	18 January 2023	Bangalore-->Mumbai

Route(Calculated Column)

1 Route = Trip_Data[Origin] &"-->"& Trip_Data[Destination]									
Trip_ID	Vehicle_ID	Driver_ID	Origin	Destination	Distance_km	Fuel_Consumed_Liters	Delivery_Status	Delivery_Date	Route
Unknown	V04	D01	Delhi	Pune	1173	108.42	On-Time	27 January 2023	Delhi-->Pune
Unknown	V06	D08	Mumbai	Bangalore	1727	161.33	On-Time	21 February 2023	Mumbai-->Bangalore
Unknown	V06	D08	Mumbai	Pune	1459	154.70	On-Time	17 February 2023	Mumbai-->Pune
Unknown	V04	D09	Hyderabad	Pune	382	26.60	On-Time	18 February 2023	Hyderabad-->Pune
Unknown	V06	D08	Pune	Mumbai	398	33.20	On-Time	15 February 2023	Pune-->Mumbai
Unknown	V06	D07	Chennai	Mumbai	1275	85.04	Late	25 February 2023	Chennai-->Mumbai
Unknown	V07	D03	Chennai	Kolkata	752	58.08	On-Time	19 January 2023	Chennai-->Kolkata
Unknown	V02	D10	Delhi	Pune	74	5.24	On-Time	01 January 2023	Delhi-->Pune
Unknown	V02	D07	Hyderabad	Hyderabad	186	16.22	On-Time	23 February 2023	Delhi-->Hyderabad
Unknown	V02	D02	Bangalore	Hyderabad	1375	105.21	Late	02 February 2023	Bangalore-->Hyderabad
Unknown	V06	D03	Kolkata	Hyderabad	419	31.17	On-Time	21 January 2023	Kolkata-->Hyderabad
Unknown	V06	D01	Kolkata	Delhi	751	51.77	On-Time	15 February 2023	Kolkata-->Delhi
T013	V05	D04	Kolkata	Chennai	1571	188.52	Late	02 February 2023	Kolkata-->Chennai
T014	V05	D05	Hyderabad	Bangalore	1524	104.51	On-Time	16 February 2023	Hyderabad-->Bangalore
T015	V05	D06	Kolkata	Mumbai	1956	179.88	On-Time	21 January 2023	Kolkata-->Mumbai
T016	V05	D06	Bangalore	Mumbai	858	92.70	Late	18 January 2023	Bangalore-->Mumbai

Visualization





Key Insights

- A total of 50 trips were completed across the fleet during the analysis period.
- Among all vehicle types (Truck, Mini Truck, and Van), Trucks recorded the highest trip count, indicating that they are the primary vehicles used for transportation operations.
- The average Fuel Efficiency is 12 km/L, showing the fleet's fuel consumption performance and helping monitor operating efficiency.
- The average Maintenance Cost per Kilometer is ₹1.42, providing visibility into vehicle maintenance expenses and cost control.
- Mumbai had the highest trip count, making it the most visited destination for deliveries.
- The fleet achieved an On-Time Delivery Rate of 60%, demonstrating satisfactory delivery performance.
- However, 40% of deliveries were delayed, indicating opportunities to improve route planning, scheduling, and operational efficiency.
- Delivery performance and vehicle utilization analysis help identify areas for improving resource allocation and reducing transportation costs.

Summary

This dashboard analyzes fleet performance and delivery efficiency using key metrics such as total trips, fuel efficiency, maintenance cost, and delivery status.

The analysis shows that trucks are the most frequently utilized vehicles, completing the highest number of trips.

The fleet maintains an average fuel efficiency of 12 km/L and a maintenance cost of ₹1.42 per kilometer.

While the organization achieved a 60% on-time delivery rate, the 40% delayed deliveries indicate the need for operational improvements.

Overall, the dashboard provides valuable insights for monitoring fleet operations, improving delivery performance, optimizing vehicle utilization, and controlling transportation costs.